

7.8 Quiz: Logarithms (Version B) — Markscheme

1. [Maximum mark: 5]

(a) $\ln 4 + \ln 7 = \ln(4 \cdot 7) = \ln 28.$ **A1**

(b) $2 \ln 5 = \ln 5^2$ **A1**

$= \ln 25.$ **A1**

(c) $-\ln \frac{1}{6} = \ln \left(\frac{1}{6} \right)^{-1}$ **A1**

$= \ln 6.$ **A1**

Accept also $-\ln \frac{1}{6} = -(\ln 1 - \ln 6) = \ln 6$, or $-\ln \frac{1}{6} = -\ln 6^{-1} = \ln 6$.

2. [Maximum mark: 5]

$\log_{10} 3 = p, \log_{10} 5 = q.$

(a) $\log_{10} 45 = \log_{10}(3^2 \cdot 5) = \log_{10} 3^2 + \log_{10} 5.$ **M1**

$= 2 \log_{10} 3 + \log_{10} 5.$ **A1**

$= 2p + q.$ **A1**

(b) Change of base: $\log_3 25 = \frac{\log_{10} 25}{\log_{10} 3} = \frac{2 \log_{10} 5}{\log_{10} 3}.$ **A1**

$= \frac{2q}{p}.$ **A1**

3. [Maximum mark: 5]

$$\log_{10} b = \frac{2}{3}, b > 0.$$

$$\text{(a)} \log_{10} \left(\frac{1}{b} \right) = -\log_{10} b. \quad \text{A1}$$

$$= -\frac{2}{3}. \quad \text{A1}$$

$$\text{(b)} \text{ Change of base: } \log_{1000} b = \frac{\log_{10} b}{\log_{10} 1000} = \frac{\log_{10} b}{3}. \quad \text{A1}$$

$$= \frac{2/3}{3} \quad \text{A1}$$

$$= \frac{2}{9}. \quad \text{A1}$$

4. [Maximum mark: 5]

$$2 \ln x = \ln 16 + 4, x = pe^q, p, q \in \mathbb{Z}^+.$$

$$2 \ln x - \ln 16 = 4. \text{ Use } m \ln x = \ln x^m: \ln x^2 - \ln 16 = 4. \quad \text{M1}$$

$$\text{Use } \ln a - \ln b = \ln \frac{a}{b}: \ln \frac{x^2}{16} = 4. \quad \text{M1}$$

$$\text{Exponentiate: } \frac{x^2}{16} = e^4 \Rightarrow x^2 = 16e^4. \quad \text{A1}$$

$$\text{Since } x > 0: x = \sqrt{16e^4} = 4e^2. \quad \text{A1}$$

$$\text{So } p = 4, q = 2. \quad \text{A1}$$

Accept any valid method, including expressing $4 = 4 \ln e$ first to combine all terms as a single log before exponentiating.